

Chart of Accounts Backend Implementation Guide

Built from the approved workbook ThinkOn_COA_V2_Renumbered_1.xlsx — the single source of truth

315 accounts	163 postable (DETAIL)	152 headers	8 categories	5 levels
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Contents

1. Source of truth & scope
 - what this document binds you to
2. Account numbering specification
 - parsing, validation, and what the code does NOT mean
3. Category model
 - the 8 top-level categories and income-statement structure
4. Database schema
 - tables, column mapping from the workbook, constraints
5. Import pipeline
 - exact steps, validations, idempotency, template cloning
6. Account flags & their runtime semantics
 - contra, control, branch-specific, clearing
7. Posting-rule seed data
 - every recurring transaction mapped to real codes
8. Database-level enforcement
 - the five triggers that are not optional
9. Reporting layer
 - trial balance, balance sheet, income statement subtotals
10. Data-validation findings in the current file
 - what import must flag today — read before coding
11. Minimal API surface
 - launch endpoints

1. Source of Truth & Scope

One workbook, one sheet, 315 rows. Everything in this document derives from it.

The approved workbook `ThinkOn_COA_V2_Renumbered_1.xlsx` (sheet شجرة الحسابات) is the authoritative Chart of Accounts, reviewed and edited by the finance team. The backend's job is to **import it faithfully, enforce its rules structurally, and never encode any account number in module code**. When this document and the workbook disagree, the workbook wins — regenerate the seed, don't patch by hand.

Current statistics: 315 accounts — 163 postable DETAIL accounts and 152 HEADER roll-up accounts, across 8 top-level categories and up to 5 hierarchy levels. Postable accounts sit at level 5 (6-digit codes) with exactly two exceptions at level 4 (see §10).

The one architectural rule: modules (Sales, Purchasing, Inventory, Payroll) never write to the GL directly and never contain a hardcoded account number. They emit accounting events; a single posting engine resolves accounts through configurable `posting_rule` rows (§7) and posts balanced journal entries. Adding an account, a branch, or a tax later is configuration, not deployment.

2. Account Numbering Specification

Hierarchical prefix numbering. The parent's code is a prefix of every child's code.

Level 1	1 digit	1	category	e.g. 1 = Assets
Level 2	2 digits	1 + 1	group	e.g. 11 = Current Assets
Level 3	3 digits	11 + 1	sub-group	e.g. 112 = Receivables
Level 4	4 digits	112 + 1	account group	e.g. 1121 = Trade Receivables
Level 5	6 digits	1121 + 2	postable account	e.g. 112101

Validation regex per level: `^[1-8]$, ^[1-8][1-9]$, ^[1-8][1-9][1-9]$, ^[1-8][1-9]{2}[1-9]$, ^[1-8][1-9]{2}[1-9][0-9]{2}$`. The 2-digit tail at level 5 allows 99 postable accounts per group; widening it later to 3 digits is non-breaking because codes are stored as text and ordering is lexicographic within a fixed level.

What the code does NOT mean: the prefix relationship is a naming convention for humans. The authoritative hierarchy is the `parent_code` column, resolved at import into `parent_account_id`. No query, report, or rollup may ever parse the code string to find a parent — import validates that prefix and parent agree (§5), and after that only the FK is used. Likewise **postability is `account_type = 'DETAIL'`, never code length**: two postable accounts currently sit at level 4 (§10).

Storage rules

- `account_code` `VARCHAR(20)` `NOT NULL`, unique per company. Never `INT` — leading semantics and future widening both die in integer columns, and Excel's date-mangling of codes is exactly the bug class text storage prevents.
- `account_level` `SMALLINT` `NOT NULL` `CHECK (account_level BETWEEN 1 AND 5)`, stored, not computed from length.
- Sort order everywhere: `ORDER BY account_code` — with fixed digits per level this yields the tree in display order for free.

3. Category Model

Eight categories, each owning one normal balance and one financial statement.

Code	Arabic	English	Normal balance	Statement
1	الأصول	Assets	Debit	Balance Sheet
2	الالتزامات	Liabilities	Credit	Balance Sheet
3	حقوق الملكية	Equity	Credit	Balance Sheet
4	الإيرادات	Revenue	Credit	Income Statement
5	تكلفة المبيعات	Cost of Goods Sold	Debit	Income Statement
6	المصاريف التشغيلية	Operating Expenses	Debit	Income Statement
7	الإيرادات الأخرى	Other Income	Credit	Income Statement
8	المصاريف الأخرى	Other Expenses	Debit	Income Statement

The category is determined by the first digit of the account code. There are **no code-range tables** (`code_range_low/high` from earlier drafts are gone) — first-digit lookup replaces them and structurally eliminates the orphan-account problem ranges had.

Income statement structure (report layer, not accounts)

```
Revenue (41) - Returns & Discounts (42) = Net Sales
Net Sales - Cost of Goods Sold (5) = Gross Profit
Gross Profit - Operating Expenses (6) = Operating Income
Operating Income + Other Income (7) - Other Expenses (81, 82) = Income Before Tax
Income Before Tax - Income Tax (83) = Net Income
```

Subtotals are computed from category/group ordering in a `financial_statement_template` config table — never posted, never stored as accounts.

4. Database Schema

Two tables carry the COA. Column mapping from the workbook is exact and total.

```
CREATE TABLE account_category (  
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  code        CHAR(1)      NOT NULL UNIQUE,      -- '1'..'8'  
  name_ar     VARCHAR(100) NOT NULL,  
  name_en     VARCHAR(100) NOT NULL,  
  normal_balance CHAR(1)   NOT NULL CHECK (normal_balance IN ('D','C')),  
  financial_statement VARCHAR(20) NOT NULL  
    CHECK (financial_statement IN ('BALANCE_SHEET','INCOME_STATEMENT')),  
  display_order SMALLINT   NOT NULL  
);  
  
CREATE TABLE gl_account (  
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  company_id   UUID NOT NULL REFERENCES company(id),  
  account_code VARCHAR(20)  NOT NULL,  
  account_name_ar VARCHAR(200) NOT NULL,  
  account_name_en VARCHAR(200) NOT NULL,  
  parent_account_id UUID REFERENCES gl_account(id),  
  category_id  UUID NOT NULL REFERENCES account_category(id),  
  account_level SMALLINT NOT NULL CHECK (account_level BETWEEN 1 AND 5),  
  account_type VARCHAR(10) NOT NULL CHECK (account_type IN ('HEADER','DETAIL')),  
  normal_balance CHAR(1) NOT NULL CHECK (normal_balance IN ('D','C')),  
  is_contra     BOOLEAN NOT NULL DEFAULT FALSE,  
  is_control_account BOOLEAN NOT NULL DEFAULT FALSE,  
  control_account_type VARCHAR(20)  
    CHECK (control_account_type IN ('AR','AP','INVENTORY')),  
  is_branch_specific BOOLEAN NOT NULL DEFAULT FALSE,  
  is_clearing     BOOLEAN NOT NULL DEFAULT FALSE,  
  is_active       BOOLEAN NOT NULL DEFAULT TRUE,  
  is_postable     BOOLEAN GENERATED ALWAYS AS  
    (account_type = 'DETAIL' AND is_active) STORED,  
  description     VARCHAR(500),  
  created_at      TIMESTAMPTZ NOT NULL DEFAULT now(),  
  updated_at      TIMESTAMPTZ NOT NULL DEFAULT now(),  
  UNIQUE (company_id, account_code),  
  CONSTRAINT chk_control CHECK  
    (is_control_account = FALSE OR control_account_type IS NOT NULL)  
);  
  
CREATE INDEX idx_gl_parent  ON gl_account(parent_account_id);  
CREATE INDEX idx_gl_company ON gl_account(company_id, is_active);  
CREATE INDEX idx_gl_control ON gl_account(company_id, control_account_type)  
  WHERE is_control_account = TRUE;
```

Workbook column → database mapping

Workbook column	Destination	Rules
account_code	gl_account.account_code	TEXT — store as string, never integer. Unique per company. 1–6 digits today; size the column VARCHAR(20) for future growth.
parent_code	resolved to gl_account.parent_account_id	Resolve to the parent row's UUID during import. This column — the code prefix — is the authoritative hierarchy.
account_name_ar	gl_account.account_name_ar	NOT NULL. Primary display name in the UI and official reports.
account_name_en	gl_account.account_name_en	NOT NULL. Secondary/exports.

Workbook column	Destination	Rules
level	gl_account.account_level	INT 1–5. Must equal LENGTH(account_code) mapped 1→1, 2→2, 3→3, 4→4, 6→5. Validate on import.
account_type	gl_account.account_type	'HEADER' 'DETAIL'. Drives is_postable.
category_code	FK → account_category	8 categories (below). Must equal the category of the first digit of account_code — validate, don't trust blindly.
normal_balance	gl_account.normal_balance	'D' 'C'. Must match the category's normal balance — validate.
financial_statement	derived via category	Do NOT store per account. It's a category attribute; storing it twice invites drift.
is_contra	gl_account.is_contra	BOOLEAN. Presentation-only: contra accounts subtract from their group in reports. Never affects posting validation.
is_control_account + control_account_type	gl_account.is_control_account / control_account_type	'AR' 'AP' 'INVENTORY'. Enforces subledger linkage at posting time (§7).
is_branch_specific	gl_account.is_branch_specific	BOOLEAN. Cash/bank accounts that get a physical instance per branch (§6.3).
is_clearing	gl_account.is_clearing	BOOLEAN. Must-net-to-zero accounts monitored by a scheduled job (§6.4).
notes	gl_account.description	Free text.

5. Import Pipeline

The workbook is imported into a template; each company gets a clone of the template.

- 1 Parse the sheet top-to-bottom, skipping fully blank rows. Trim all strings.
- 2 Normalize booleans: 'True'/'TRUE'/'true' → TRUE, else FALSE.
- 3 Validate per row:
 - a. code matches the level regex (\$2); level matches code length (1,2,3,4,6 → 1..5)
 - b. parent_code exists earlier in the file (single forward pass suffices – the sheet is in tree order)
 - c. code starts with parent_code (prefix consistency)
 - d. normal_balance == category(first digit).normal_balance
 - e. financial_statement column == category's statement (then DISCARD the column)
 - f. control_account_type present iff is_control_account
- 4 Validate globally:
 - a. codes unique
 - b. every HEADER has ≥1 child; every DETAIL has 0 children
 - c. exactly the expected root set {1..8}
- 5 On ANY validation failure: import nothing, emit the full findings list (§10 format). Never partial-import a chart of accounts.
- 6 Write to coa_template + coa_template_account (company-independent master copy).
- 7 Company provisioning clones the template into gl_account rows for that company, with every account is_active = TRUE by default; the activation screen then deactivates what the company's business doesn't need. Deactivation, never deletion.
- 8 Idempotency: re-importing the same workbook version is a no-op; importing a new version diffs against the template – new codes are added, changed names are updated, removed codes are flagged (never auto-deleted if any company activated them). Code, category, level, and normal_balance of an existing account are immutable through re-import once any company has posted to it.

Step 5 is the point of the whole pipeline. The current workbook does not pass validation 4b — see §10. That is by design: the import tool telling the finance team exactly what to fix in the sheet is the workflow, not an obstacle to it.

6. Account Flags & Runtime Semantics

Four boolean flags in the workbook each drive a specific backend behavior.

6.1 is_contra — 17 accounts

Presentation only: in any grouped report, a contra account's balance subtracts from its group total (allowances, accumulated depreciation, returns, discounts earned/allowed, dividends, treasury shares). It never changes posting validation and is never inferred from names. Current set: 113301, 114501, 122101, 122102, 122103, 122104, 122105, 122106, 123201, 124102, 341101, 341102, 341103, 421101, 421102, 421103, 532104.

6.2 is_control_account — 10 accounts

Code	Account	Type
112101	ذمم أطراف ذات علاقة - تحكم	AR
114101	مواد خام	INVENTORY
114102	قطع غيار ومواد تعبئة	INVENTORY
114201	إنتاج تحت التشغيل	INVENTORY
114202	أعمال خدمية/مشاريع تحت التنفيذ	INVENTORY
114301	بضاعة جاهزة	INVENTORY
114302	بضاعة لدى الغير (أمانات)	INVENTORY
114401	بضاعة بالطريق	INVENTORY
211101	ذمم الموردين - حساب تحكم	AP
211102	ذمم أطراف ذات علاقة - تحكم	AP

Runtime rule: a journal line posting to an AR control must carry `customer_id`; AP control → `vendor_id`; INVENTORY control → `item_id`/warehouse reference. Enforced by trigger (§8). Control accounts also reject manual journal entries entirely — they are fed only by the posting engine — and each control account's GL balance must continuously equal the sum of its subledger (nightly reconciliation job; drift = a code path bypassed the engine).

Before wiring AR: note the §10 blocker — the trade-AR account (1121) is not currently flagged as a control account; only related-party AR (112101) is. Do not code around this; the sheet must be fixed.

6.3 is_branch_specific — 4 accounts

Code	Account
111101	الصندوق الرئيسي
111102	صندوق نقطة البيع (كاشير)
111103	المصروفات الثرية
111201	حساب بنكي جاري

These are physical cash/bank accounts: each branch gets its own instance. Implementation: keep ONE `gl_account` row and require `branch_id` on every journal line touching it (branch as a dimension), plus a `bank_account/cash_till` operational table mapping physical tills and bank accounts to (`gl_account_id`, `branch_id`). Do NOT clone `gl_account` rows per branch — that

reintroduces the account-explosion problem the control-account design exists to prevent. Everything not flagged branch-specific still accepts `branch_id` as an optional reporting dimension.

6.4 `is_clearing` — 6 accounts

Code	Account
111401	مبالغ تحت التحصيل
117101	حساب وسيط بين الفروع
117102	حساب تسوية معلق
212101	بضاعة مستلمة غير مفوترة (GRNI)
214102	صافي الضريبة المستحقة للتسوية
823101	فروقات تقريب العملة

Must-net-to-zero accounts. A scheduled job (daily, and hard-gate at period close) checks each clearing account's balance per branch; any balance above a configurable threshold raises a notification and blocks period close until acknowledged with a reason code. A material balance in 823101 (rounding) is a software bug, not an accounting event — alert engineering, not just finance.

7. Posting-Rule Seed Data

Each row becomes one `posting_rule` + its line templates. Module code references rule codes, never account codes.

Event	rule_code	Debit	Credit
Sales invoice (credit)	AR_INVOICE	1121 Trade Receivables [customer_id]	411xxx–414xxx Revenue + 214101 Output Tax
COGS at shipment	INV_COGS	511101/02/03 COGS	1143xx Finished Goods
Customer payment (cheque/cash)	AR_PAYMENT	111401 Undeposited / PDC	1121 Trade Receivables [customer_id]
Bank deposit of receipts	BANK_DEPOSIT	111201 Bank – Current	111401 Undeposited / PDC
Sales return	AR_CREDIT_MEMO	421101 Sales Returns	1121 Trade Receivables [customer_id]
Goods receipt before bill	INV_RECEIPT	1141xx–1144xx Inventory	212101 GRNI
Vendor bill matched to receipt	AP_BILL	212101 GRNI (+531201 PPV on cost diff)	211101 AP Control [vendor_id]
Vendor payment with WHT	AP_PAYMENT_WHT	211101 AP Control [vendor_id]	111201 Bank + 214202 Withholding Tax
Employee advance	EMP_ADVANCE	115101 Employee Advances*	111201 Bank
Monthly payroll accrual	PAYROLL_ACCRUE	621xxx Salaries & Benefits	213101 Salaries Payable + 213201 Social Security
Monthly depreciation	FA_DEPRECIATE	641101 Depreciation Expense	1221xx Accumulated Depreciation
Fixed asset disposal	FA_DISPOSE	111201 Bank + 1221xx Accum. Dep.	121xxx Asset at cost + 711102 Gain / 821101 Loss
Inter-branch transfer (in transit)	BRANCH_TRANSFER	117101 Inter-Branch Clearing	Inventory/Cash (source branch) — then reversed on receipt
Period-end FX revaluation	FX_REVAL	AR/AP control or cash balance	712102 Unrealized FX Gain / 822102 Unrealized FX Loss
Periodic tax settlement	TAX_REMIT	214101 Output Tax	116101 Input Tax* + 111201 Bank (net)
Doubtful-debt provision	AR_PROVISION	651101 ECL Expense	113301 Allowance (ECL)
Year-end close	YEAR_END_CLOSE	All income-statement accounts	331102 Current Year Earnings

Rows marked * reference accounts that must be verified against the sheet at seed time (§10, INFO findings). The seeder must resolve every referenced code against `gl_account` and fail loudly on a miss — never fall back to a guessed or nearest account.

Timing rule: revenue and its matching COGS must post in the same transaction boundary. If shipment and invoicing can straddle a reporting cutoff, the flow needs the deferred-COGS pattern; do not let gross margin float for days between the two events.

8. Database-Level Enforcement

Five rules enforced by the database itself. Application-layer checks are a courtesy copy, not the enforcement.

T1	Postability	BEFORE INSERT on journal_line: reject if the account's is_postable is FALSE (headers, deactivated accounts).
T2	Balance	DEFERRABLE INITIALLY DEFERRED constraint trigger at commit: SUM(debit) = SUM(credit) per journal_entry. An unbalanced entry can never exist, even via a bug or a migration script.
T3	Immutability	BEFORE UPDATE/DELETE on journal_entry + journal_line: RAISE EXCEPTION when status = 'POSTED'. Corrections are reversing entries, never edits.
T4	Control linkage	BEFORE INSERT on journal_line: if the account is a control account, the matching subledger reference (customer_id / vendor_id / item ref) must be non-null.
T5	COA audit	AFTER UPDATE on gl_account: write old/new values to audit_log via trigger, not application code. Once an account has any posted line, account_code, category_id, account_level, and normal_balance become immutable (enforced in the same trigger); only names, description, and is_active stay editable.

Journal line design (unchanged from the agreed engine spec): separate non-negative debit_amount/credit_amount columns with a one-sided CHECK, plus debit_amount_base/credit_amount_base and branch_id present from day one even while single-currency/single-branch — adding columns to an empty table is free; adding them to millions of posted lines is a migration.

9. Reporting Layer

All reports read one view; hierarchy rollup is one recursive CTE.

```
CREATE VIEW gl_posted_lines AS
SELECT jl.*, je.entry_date, je.fiscal_period_id, je.company_id
FROM journal_line jl
JOIN journal_entry je ON je.id = jl.journal_entry_id
WHERE je.status = 'POSTED';

-- Rollup of any account including all descendants:
WITH RECURSIVE sub AS (
  SELECT id FROM gl_account WHERE id = :account_id
  UNION ALL
  SELECT g.id FROM gl_account g JOIN sub s ON g.parent_account_id = s.id
)
SELECT SUM(jl.debit_amount - jl.credit_amount)
FROM gl_posted_lines jl WHERE jl.account_id IN (SELECT id FROM sub);
```

- **Trial balance** groups postable accounts by code order; hard-fails (never renders) if total debits $\neq$ total credits — that state means corruption, not data.
- **Balance sheet**: categories 1–3, cumulative since inception, as-of date bound.
- **Income statement**: categories 4–8, period-bound, with the §3 subtotal template; contra accounts subtract within their group (net sales = 41 – 42).
- **Branch P&L** = the same income statement view with `WHERE branch_id = :b` — no second pipeline.
- Balances served from the period-cache table (`account_balance`), updated in the same transaction as each post; live aggregation only inside the open period.

10. Data-Validation Findings in the Current File

What the §5 import validator reports against ThinkOn_COA_V2_Renumbered_1.xlsx today. BLOCKERs stop import; the finance team fixes them in the sheet, not engineering in the database.

Severity	Finding
BLOCKER	1121 is typed DETAIL but has a child (112101) The invariant "DETAIL accounts have no children" is violated in exactly one place. Two valid fixes, to be chosen by the accountants: (a) retype 1121 as HEADER and add a new detail child 112102 "ذمم العملاء - محلي" as the trade-AR posting account, or (b) move 112101 up to be a sibling (1123) and keep 1121 as a childless DETAIL. Import must hard-fail until one is applied.
BLOCKER	No trade-AR control account is flagged The only account flagged control_account_type='AR' is 112101 (related-party AR). The main trade receivables account (1121) has is_control_account=FALSE. As it stands, customer invoices would post to an account with no enforced customer_id linkage — aging reports break silently. Flag the trade AR posting account as an AR control before go-live.
WARNING	2112 أوراق الدفع is a HEADER with no children A header leaf can never receive postings and never roll anything up. Either add its detail accounts (post-dated cheques payable was removed in editing) or retype it DETAIL.
WARNING	Two postable accounts sit at level 4 (1121, 1122) Everywhere else, postable accounts are level-5. This is legal under the rule "postability = DETAIL, not level", but the UI and any code that assumes 6-digit postable codes will be wrong. Either normalize to level 5 for consistency, or make sure no code ever infers postability from code length.
WARNING	Bank clearing headers 1113/1114 carry editing artifacts 1113 is named "نقد تحت التحصيل والتسوية" / Bank Account - FX Clearing" but its child is the FX bank account itself; 111401 is named "مبالغ تحت التحصيل" / PDC Cheques" mixing undeposited funds with post-dated cheques. Functional, but the names will confuse the first accountant who reconciles a bank. Recommend a naming pass.
INFO	No Input-Tax (VAT recoverable) asset account exists 214102 "صافي الضريبة المستحقة للتسوية" exists on the liability side, but the 116xxx input-tax asset from the earlier design was removed in editing. Fine for a non-recoverable regime; the tax engine's input_tax_account_id stays NULL until the account is added.
INFO	No employee-advance / prepaid asset codes referenced by two posting rules EMP_ADVANCE and TAX_REMIT reference accounts (115101 employee advances, 116101 input tax) that may not exist under those codes in the final sheet — the import tool must resolve posting-rule account references against the actual sheet and fail loudly on any miss, never guess.

11. Minimal API Surface

GET	/companies/{id}/gl-accounts	?category=&type=&active=&q=
GET	/companies/{id}/gl-accounts/{id}/hierarchy	ancestors + descendants
POST	/companies/{id}/gl-accounts	create (validates §2 + §5 rules)
POST	/companies/{id}/gl-accounts/{id}/deactivate	never DELETE
POST	/companies/{id}/journal-entries	DRAFT / PENDING_APPROVAL
POST	/companies/{id}/journal-entries/{id}/approve	approver ≠ creator, server-side
POST	/companies/{id}/journal-entries/{id}/reverse	
GET	/companies/{id}/reports/trial-balance	?asOfDate=
GET	/companies/{id}/reports/balance-sheet	?asOfDate=
GET	/companies/{id}/reports/income-statement	?periodStart=&periodEnd=&branchId=
GET	/companies/{id}/clearing-accounts/status	§6.4 monitor
POST	/admin/coa-template/import	§5 pipeline; returns findings list

No PUT/PATCH/DELETE on a POSTED journal entry, ever — 409 Conflict if attempted. That is trigger T3 made visible in the API.

Definition of done for Phase 0/1: the workbook imports with zero findings after the finance team applies the §10 fixes; all 8 category rollups reconcile via the recursive CTE; a manual journal entry posts, balances, appears on the trial balance, and a direct SQL UPDATE against it is rejected by the database itself.